

Worksheet — Springs in Series and Parallel

Simulation 06 · use with *Simulation - Springs in series and parallel.html* · extension of Hooke's law (beyond the IB SL guide)

Name: _____ Class: _____ Date: _____

How to use this worksheet

Open **Simulation - Springs in series and parallel.html** in a browser (it works offline). Choose the arrangement and the two springs, add a load and press ● **Record**. The **Tasks** tab has the same three parts as this sheet.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab with two school springs (≈ 10 min) · **Part 3** homework (≈ 30 min).

Use $g = 9.81 \text{ N kg}^{-1}$. Effective spring constant $k_{\text{eff}} = F/x$, with x the total extension in metres.

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. Two identical springs in series carry the same load as one spring. Total extension?			
2. The same two springs in parallel carry the same load. Extension?			
3. In series with 200 g (1.96 N), what force acts in the top spring?			
4. Soft (12 N m^{-1}) and stiff (40 N m^{-1}) springs in parallel share a load. Which pulls harder?			

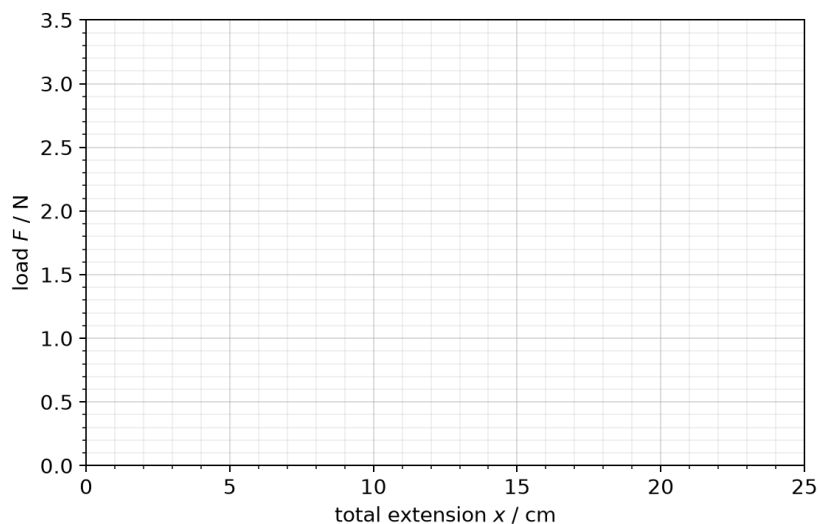
5. Show that for two springs in series $1/k_{\text{eff}} = 1/k_A + 1/k_B$, and for two springs in parallel $k_{\text{eff}} = k_A + k_B$. (Hint: in series the force is the same; in parallel the extension is the same.)

Working

Part 2 — During the lab: two school springs

load / g	F / N	one spring x / cm	series x / cm	parallel x / cm
100				
200				
300				

6. Plot all three sets on one graph and draw a best-fit line through the origin for each.



7. k (one spring) = _____ k (series) = _____ k (parallel) = _____ N m^{-1} . Which rule do your results support?

Part 3 — Homework

8. Soft (12 N m^{-1}) and stiff (40 N m^{-1}) springs in **series**: calculate k_{eff} and the total extension with 200 g hanging.

Working

9. The same springs in **parallel** with 200 g: calculate k_{eff} , the common extension and the force in each spring. Draw the free-body diagram of the load.

Working

10. A 1200 kg car rests on 4 identical springs, each compressed by 3.0 cm. Calculate the spring constant of one spring and the effective spring constant of all four together.

Working

11. A rubber band's F - x graph curves. Why can't you give it a single spring constant?
